

Question	Marking Guidance	Mark	Comments
05.1	1. (Hydrolase/ATP hydrolysis) releases <u>energy</u> ; 2. (For) active transport (by X) of H^+ in(to vacuole); 3. (Causes) <u>co-transport</u> (by Y) of H^+ out and fructose in; 4. Decreases water potential so water moves in(to vacuole) by osmosis;	4 (4 x AO2)	1. Reject make/create/produce energy 2 and 3 Reject hydrogen once 2 and 4 Accept 'cell' for 'vacuole' 4. Accept more negative water potential for 'decreases water potential' 4. Accept Ψ for 'water potential', but ignore WP

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05.2	1. No/less ATP (produced/available); 2. (So) no/less ATP hydrolysed OR (So) no/less energy released/transferred OR (So) ATP hydrolase/enzyme inactive/less active; 3. Less/no H^+ enters (vacuole) so pH increases OR H^+ (still co-) transported out (of vacuole) so pH increases;	3 (3 x AO2)	2. Reject no energy produced 3. Accept 'cell' for 'vacuole' 3. Accept acidity decreased OR becomes neutral OR becomes alkali OR a relevant change in pH value (eg from pH 4 to pH 5) 3. Ignore method of H^+ transport

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05.3	1. Increased/more (gene) expression/transcription (produces) more carrier (proteins)/X/Y; 2. (So) more/faster active transport H^+ (in)	2 (2 x AO2)	1. Accept "it increases" for 'increased (gene) expression'

	OR (So) more/faster co-transport OR (So) more/faster H ⁺ /fructose movement OR		2. Reject fructose moving out
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